

DDoS attack detection using CNN-BiLSTM with attention mechanism

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ABSTRACT

In today's interconnected world, nearly every aspect of modern life relies on internet-based services. However, the widespread adoption of these services has led to a significant increase in cyberattacks, particularly Distributed Denial of Service (DDoS) attacks—which pose a major threat to the stability of critical infrastructure. Therefore, timely and accurate detection of DDoS attacks is essential for cybersecurity. Although deep learning has shown promise in intrusion detection, existing systems often struggle with multiclass DDoS classification, class imbalance, and adaptability to evolving traffic patterns—especially in resource-constrained IoT and IIoT environments. To address these challenges, this study proposes a novel lightweight hybrid deep learning model that integrates Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM) networks, and an attention mechanism to enhance feature extraction and improve classification performance. The model is evaluated on two benchmark datasets, CICDDoS2019 and Edge-IIoT, achieving classification accuracies of 99.78% and 98.84%, and ROC-AUC scores of 100% and 99.98%, respectively. Comparative analysis with baseline models and prior studies highlights the model's robustness, efficiency, and suitability for real-time deployment in diverse and constrained environments.

1. Introduction

In this digital era, protecting private information using traditional security solutions has become increasingly complex. The rise in cyberattacks, especially Distributed Denial of Service (DDoS) attacks, poses serious threats to network infrastructures. These attacks can disrupt essential services and result in significant financial and reputational damage [1]. As a result, developing accurate and efficient mechanisms for detecting and classifying DDoS attacks is critical for maintaining network reliability and security.

Despite advances in intrusion detection systems (IDS), existing solutions still face critical challenges in effectively handling DDoS attacks. These include difficulties in multiclass classification, sensitivity to class imbalance, limited generalization to evolving attack patterns, and constraints in real-time deployment—particularly in resource-limited environments such as IoT and IIoT networks. To address these issues, this study proposes a novel lightweight hybrid deep learning model that integrates Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM) networks, and attention mechanisms. This architecture aims to enhance feature extraction, improve classification performance, and enable robust detection of diverse DDoS attack types.

Recent research demonstrates the potential of machine learning and deep learning approaches in identifying malicious network traffic.

While traditional machine learning methods rely heavily on manual feature engineering, deep learning models inherently support feature extraction and classification within their architecture [2–4]. Deep learning methods – especially those using attention mechanisms – have shown greater adaptability by focusing on relevant features while ignoring irrelevant input, which is vital for identifying new or evolving DDoS variants [5,6].

Research questions

This study is driven by the following key research questions:

- RQ1:** How can a hybrid deep learning architecture combining CNN, BiLSTM, and attention mechanisms enhance the accuracy of DDoS attack detection and classification?
- RQ2:** How effectively does the proposed model handle multiclass DDoS classification challenges, including class imbalance and dynamic traffic patterns?
- RQ3:** How does the proposed approach perform compared to existing models in terms of detection accuracy across different DDoS attack types and datasets?

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4. **RQ4:** Can the proposed model be optimized to remain lightweight and computationally efficient for real-time deployment in resource-constrained environments such as IoT and IIoT networks?

To validate the proposed approach, we conduct experiments using two publicly available and comprehensive datasets: CICDDoS2019 and Edge-IIoT. These datasets represent diverse attack types, varying network conditions, and data imbalances. These datasets were selected for their complementary strengths—CICDDoS2019 delivers in-depth, flow-based insights into DDoS attacks within traditional network infrastructures, while Edge-IIoT captures a more contemporary and realistic spectrum of attack scenarios tailored to the constraints and complexities of IoT and IIoT environments. The results highlight the model's ability to maintain high performance across these challenges.

The main contributions of this study are as follows:

- We propose a novel hybrid architecture that combines CNN and BiLSTM networks with an attention mechanism to capture both spatial and temporal features effectively.
- The attention mechanism improves classification accuracy by dynamically focusing on salient features and adapting to new attack patterns.
- Our model is lightweight, with 5052 and 5105 trainable parameters for CICDDoS2019 and Edge-IIoT, respectively, demonstrating efficiency without sacrificing performance.
- The proposed model achieves classification accuracies of 99.78% and 98.84%, and ROC AUC scores of 100% and 99.98% on the CICDDoS2019 and Edge-IIoT datasets, respectively.

The rest of the paper is structured as follows: Section 2 reviews deep learning studies on DDoS attacks, Section 3 provides an explanation of the dataset description and preprocessing techniques, Section 4 presents our proposed methodology, Section 5 offers an analysis of experiments and their results, and Section 6 concludes the study while discussing avenues for future research.

2. Related work

The detection of Distributed Denial of Service (DDoS) attacks has been extensively studied, with a wide array of techniques proposed to improve both the accuracy and efficiency of detection systems. In recent years, deep learning (DL) approaches have gained significant traction due to their ability to automatically extract complex features from network traffic. These methods are now employed across various domains, including healthcare, vehicle design, manufacturing, and cybersecurity [7]. This section provides an overview of recent research efforts, with a focus on the application of deep learning for DDoS attack detection.

In the context of Industrial IoT (IIoT), Muna et al. [8] proposed an anomaly detection system (ADS) based on the UDAE deep learning algorithm. The system was trained on typical activity patterns to identify potential cyber threats. Tested on the NSL-KDD and UNSW-NB15 datasets, it achieved an accuracy of 99% with a false positive rate of 1.8%. However, the model's training time was relatively long – averaging 194.37 s per epoch on NSL-KDD and 227.30 s on UNSW-NB15 – and the datasets may lack coverage of contemporary DDoS attacks.

Parra et al. [9] proposed a distributed architecture combining a Distributed Convolutional Neural Network (DCNN) for edge-level IoT micro-security and a Long Short-Term Memory (LSTM) network at the server backend. Evaluated using the N_BaIoT dataset, their system achieved 94.30% accuracy in multiclass classification and 94.8% in binary classification. The results were promising but indicated potential for further accuracy improvements. Akgun et al. [10] leveraged various deep learning architectures, including Inception modules, DNNs, CNNs, and LSTMs, to build an intrusion detection system. Their model

attained 99.9% accuracy for binary classification and 99.3% for multiclass classification, showcasing the benefits of using inception-like Conv1D layers.

Cil et al. [11] also applied deep learning to the CICDDoS2019 dataset using a DNN model. They reported an F1-score of 99.98%, accuracy of 99.97%, precision of 99.99%, and recall of 99.98% on Dataset 1. However, results on Dataset 2 showed slightly lower performance, highlighting challenges in multiclass DDoS detection. Ferrag et al. [12] conducted a detailed survey of deep learning techniques for intrusion detection, categorizing cybersecurity datasets into seven groups and analyzing CNN, RNN, and DNN models. While CNNs showed strong performance, they were less effective at capturing long-term dependencies in IoT traffic. Assis et al. [13] developed a defense system for SDN environments that used Gated Recurrent Units (GRU), achieving 97.09% accuracy across different conditions using the CICDDoS2019 and CICIDS2018 datasets.

Wei et al. [14] proposed a hybrid approach combining an autoencoder for feature extraction with a multilayer perceptron for DDoS classification. Their method achieved over 98% accuracy and F1-score on the CICDDoS2019 dataset. Despite promising results, the model's generalizability to diverse network environments and attack types remains to be validated. In another notable work, the authors of [15] introduced DDoS-FOCUS, a mitigation strategy for fog nodes based on a CNN-BiLSTM hybrid model. The system achieved a detection accuracy of 99.8% during preliminary testing. Najar et al. [16] proposed an advanced feature selection approach for robust DDoS detection. Their model achieved 96.82% accuracy, 96.76% precision, and a 0.189 ms prediction time on the CICDDoS2019 dataset. The combination of pre-processing and feature selection improved classification performance over existing baselines. Another study [17] introduced a computationally efficient CNN-based model that incorporated random sampling and techniques like low-information gain and PCA for feature selection. The model achieved 99.99% accuracy in binary classification and 98.44% in multiclass classification, outperforming several existing approaches. Bala et al. [18] conducted a systematic review of AI-based techniques for IoT DDoS detection, analyzing studies from 2019 to 2023. Their review covered IoT traffic characteristics, botnet lifecycles, attack taxonomies, datasets, AI tools, and identified research gaps, offering direction for future model development. Nuhu et al. [19] presented a literature review focused on deep learning and feature fusion techniques for detecting DDoS attacks in IoT networks. The study emphasized the effectiveness of combining features from multiple sources to improve detection accuracy and reviewed trends, applications, and limitations, highlighting the need for continued innovation to address evolving threats in IoT security.

Table 1 provides a summary of prominent studies in the literature, highlighting their methodologies, dataset descriptions, and corresponding accuracy and time metrics. These studies collectively demonstrate the effectiveness of deep learning techniques in DDoS attack detection, offering valuable insights into the performance and capabilities of various models. While many approaches achieve high accuracy, they often suffer from key limitations: reliance on outdated or narrow datasets (e.g., NSL-KDD), limited performance in multi-class detection, lack of adaptability to heterogeneous traffic patterns, and insufficient efficiency for deployment in real-time, resource-constrained environments. Furthermore, many methods overlook the benefits of contextual or sequential feature learning. The proposed CNN-BiLSTM-AT model addresses these gaps by integrating spatial (CNN), temporal (BiLSTM), and contextual (attention) learning, enabling it to generalize effectively across modern (CICDDoS2019) and real-world industrial (Edge-IIoT) datasets. This architecture not only achieves superior detection accuracy but also demonstrates robust performance across diverse attack types and network conditions, making it suitable for practical, real-time deployment in evolving threat landscapes.

Table 1
Summary of Related Studies on Deep Learning-Based DDoS Intrusion Detection System.

Study[Ref]	Methodology	Dataset	Results	Time
[8]	Anomaly detection system (ADS) for cyber threats in industrial IoT using deep learning (UDAE algorithm).	NSL-KDD, UNSW-NB15	Detection: 99%, FPR: 1.8%	2.25s 5.55s
[9]	Distributed architecture employing DCNN and LSTM networks for detecting phishing and botnet attacks.	N_BaIoT	Multiclass: 94.30%, Binary: 94.8%	*
[10]	Inception architecture with DNN, CNN, and LSTM.	CICDDoS2019	Binary: 99.9%, Multi-class: 99.3%	7s, 11 s, and 35 s per epoch
[11]	DNN (Deep Neural Network)	CICDDoS2019	Dataset1 99.97% and Dataset2 94%	27.01 μ s (Prediction time)
[12]	DNN, RNN, CNN	CICDDoS2018 and BoT-ToT	97.28%, 97.31%,97.38% (CICDDoS2018) and 98.22%,98.31%, 98.37% (BoT-ToT)	219.29 s (Inference time)
[13]	Gated Recurrent Units (GRU)	CICDDoS2019 and CICIDS2018	97.28%, 97.31%,97.38% (CICDDoS2018) and 97.09%	*
[14]	AE-MLP	CICDDoS2019	98.34%	*
Proposed Model	CNN-BiLSTM-AT	CICDDoS2019, Edge IIoT	99.78%, 98.84%	441.61 s, 48.33 s

Table 2
Distribution of the CICDDoS2019 dataset before and after data preprocessing.

Phase	Class	Instances	Duplicate Instances
Before-Preprocessing	Normal	56,863	NA
	Attack	50 063 107	NA
Total	50119970		NA
After-Preprocessing	Normal	50,549	6,314
	Attack	8 180 084	41883023
Total	8230633		41889337

3. Data description and preprocessing techniques

In this study, we utilize two distinct datasets, namely ‘CICDDoS2019 [20]’ and ‘Edge IIoT [21],’ which undergo a rigorous data cleaning process as a preliminary step. These datasets were selected due to their complementary characteristics—CICDDoS2019 offers detailed, flow-based DDoS attack records in traditional network environments, while Edge-IIoT provides a modern, realistic view of diverse attack scenarios in resource-constrained IoT and IIoT ecosystems. Together, they offer a comprehensive and diverse testing ground for assessing both multiclass classification performance and real-world applicability across different network settings.

3.1. CICDDoS2019 dataset

The study utilized the widely recognized CIC-DDoS2019 dataset [20], collected in real-world environments. This dataset comprises seven distinct DDoS attacks recorded on the first day and twelve attack types observed on the second day. It consists of 11 CSV files with 13 labels, encompassing attack categories and normal packets, with each instance featuring 88 attributes. The WebDDoS attack was excluded due to its infrequent occurrence. The dataset is accessible in both PCAP and CSV file formats. Table 2 and Fig. 4(A) show the dataset distribution before and after preprocessing, respectively, along with the dropped columns during data preprocessing. Additionally, Fig. 1 displays the pie chart of class distribution for both binary and multiclass categories of the CICDDoS2019 dataset. Furthermore, in the pre-processing stage, we eliminated non-essential features listed in Table 3 from the Edge IIoT dataset, as they did not play a crucial role in our analysis.

Table 3
List of Dropped Features from CICDDoS2019 Dataset during Data Preprocessing.

Features	Features
Unnamed: 0	Flow ID
Timestamp	Source IP
Source Port	Destination IP
Destination Port	SimilarHTTP
Bwd PSH Flags	Fwd URG Flags
Bwd URG Flags	FIN Flag Count
PSH Flag Count	ECE Flag Count
Fwd Avg Bytes/Bulk	Fwd Avg Packets/Bulk
Fwd Avg Bulk Rate	Bwd Avg Bytes/Bulk
Bwd Avg Packets/Bulk	Bwd Avg Bulk Rate
Subflow Fwd Packets	Subflow Bwd Packets
Subflow Fwd Bytes	Subflow Bwd Bytes
RST Flag Count	Fwd Header Length.1
Fwd PSH Flags	Bwd Packet Length Std
Bwd Packet Length Mean	URG Flag Count
Bwd Packet Length Min	SYN Flag Count
Inbound	Bwd Packet Length Max
Avg Bwd Segment Size	Bwd IAT Std
Down/Up Ratio	CWE Flag Count
Total Length of Bwd Packets	Protocol
Flow Duration	Fwd IAT Total
Min Packet Length	ACK Flag Count
min seg size forward	

3.2. Edge IIoT dataset

The Edge IIoT dataset, introduced by Ferrag et al. [21] in 2022, is a comprehensive cybersecurity dataset for IoT and IIoT applications. It includes 63 features, covering 15 attack types, including benign ones, providing a realistic representation of IoT threats. The 15 attack types are denial-of-service, distributed denial-of-service, information collecting, injection, man-in-the-middle, and malware. The dataset comprises 2,219,201 attacks, with 1,615,643 regular instances and 603,558 attack instances. To ensure balanced class distribution, a stratification approach was used, with 20% for testing and 80% for training. Fig. 2 displays the pie chart of class distribution for both binary and multiclass categories of the Edge IIoT dataset. Additionally, A detailed breakdown of the Edge IIoT dataset instances can be found in Table 4 and Fig. 4(B). Also, during the pre-processing stage, irrelevant features listed in Table 5 were removed from the Edge IIoT dataset as they were not crucial for our analysis.

As depicted in Figs. 1 and 2, the datasets showed class imbalance, with most instances belonging to the normal class. To counter this

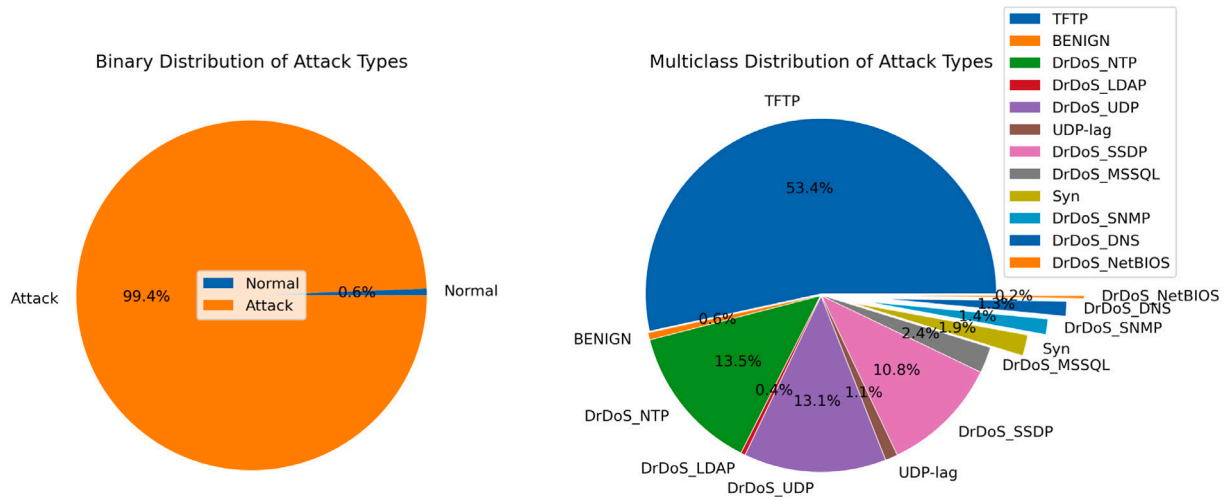


Fig. 1. Distribution of Records Before and After Data Preprocessing in CICDDoS2019 Dataset.

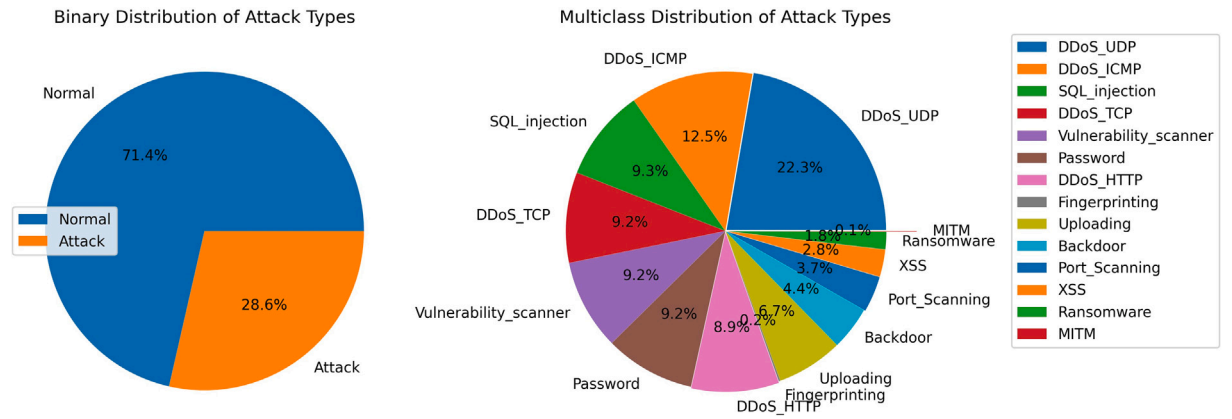


Fig. 2. Distribution of Records Before and After Data Preprocessing in Edge IIoT dataset.

Table 4

Distribution of the Edge IIoT dataset before and after data preprocessing.

Phase	Class	Instances	Duplicate Instances
Before-Preprocessing	Normal	1,615,643	NA
	Attack	603,558	NA
Total	2219201		NA
After-Preprocessing	Normal	1,363,997	251,646
	Attack	545,662	57,896
Total	1909659		309542

bias during training, we adopted the weighted random sampling technique, effectively mitigating the imbalance issue and ensuring a more balanced representation of the data during training.

To further illustrate the relationships among the selected features, we have included a correlation heatmap (see Fig. 3). This visual representation highlights the correlations between features, providing insights into how different features interact and contribute to the classification task. The heatmap helps to identify highly correlated features that may carry redundant information, as well as features that independently contribute to the model's performance.

3.3. Data preprocessing

To develop a robust classification model for detecting DDoS intrusions, we conducted comprehensive preprocessing on the initial

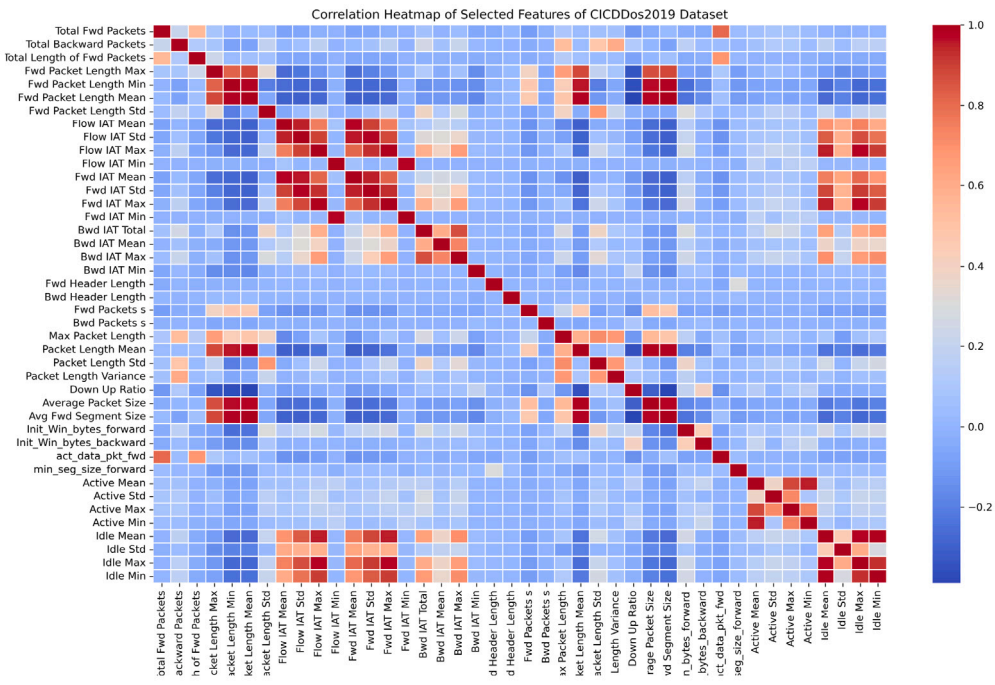
Table 5

List of Dropped Features from Edge IIoT Dataset during Data Preprocessing.

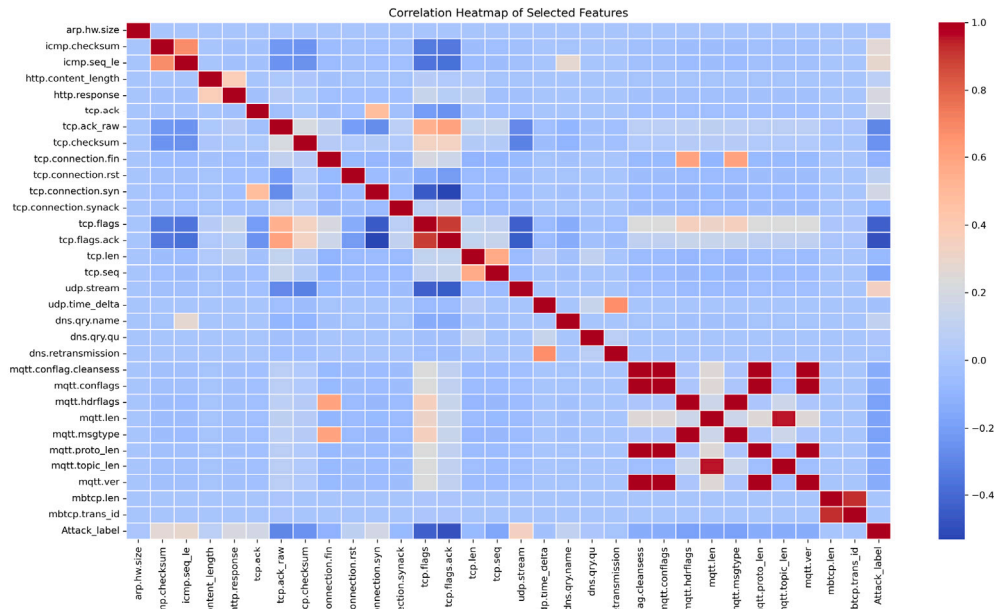
Dropped Features	Low Variance Features
frame.time	arp.opcode
ip.src_host	icmp.unused
ip.dst_host	http.tls.port
arp.src.proto_ipv4	dns.qry.type
arp.dst.proto_ipv4	dns.retransmit_request
http.file_data	dns.retransmit_request_in
http.request.full_uri	mqtt.msg_decoded_as
icmp.transmit.timestamp	mbtcp.unit_id
http.request.uri.query	tcp.options
tcp.payload	tcp.srcport
tcp.dstport	udp.port
mqtt.msg	

datasets to address issues such as missing values, erroneous data, misfits, and distortions, which could potentially impact data accuracy and reliability. We employed various techniques including imputing missing data with consistent values, and removing or replacing special symbols and corrupted codes.

Feature Selection: Feature selection was a critical step in our preprocessing pipeline to optimize model performance. We eliminated features with null values, duplicate entries, low information gain, or quasi-constant values. This process was guided by statistical analyses to ensure that only the most relevant features were retained, thereby enhancing the model's capacity to detect DDoS attacks.



(a) Correlation heatmap of selected features of CICDDoS2019 Dataset



(b) Correlation heatmap of selected features of Edge IIoT Dataset

Fig. 3. Correlation heatmaps of selected features.

Tables 3 and 5 list the features removed from the CICDDoS2019 and EDGE IIoT datasets, respectively. The selection criteria for dropping these features were based on their low variance, high correlation with other features, or minimal contribution to the model's predictive power. For example, features like `ip.src_host` and `packets_length` were evaluated based on their statistical properties and deemed redundant or minimally informative for our specific detection goals.

Initially, the CICDDoS2019 dataset contained 88 features, which were reduced to 43 after preprocessing. Similarly, the EDGE IIoT dataset was streamlined from 63 features to 40. This reduction was

achieved by removing features that either did not vary significantly across samples or were highly correlated with other features, ensuring the model focuses on the most informative data.

Handling Class Imbalance: Class imbalance was identified in both datasets, posing a significant risk of compromising classification accuracy and leading to overfitting. In a class-imbalanced dataset, the model tends to be biased towards the majority class, thus underperforming on minority classes which are often the classes of interest in intrusion detection.

To mitigate this issue, we employed the weighted random sampler technique. This approach ensures that each class is represented

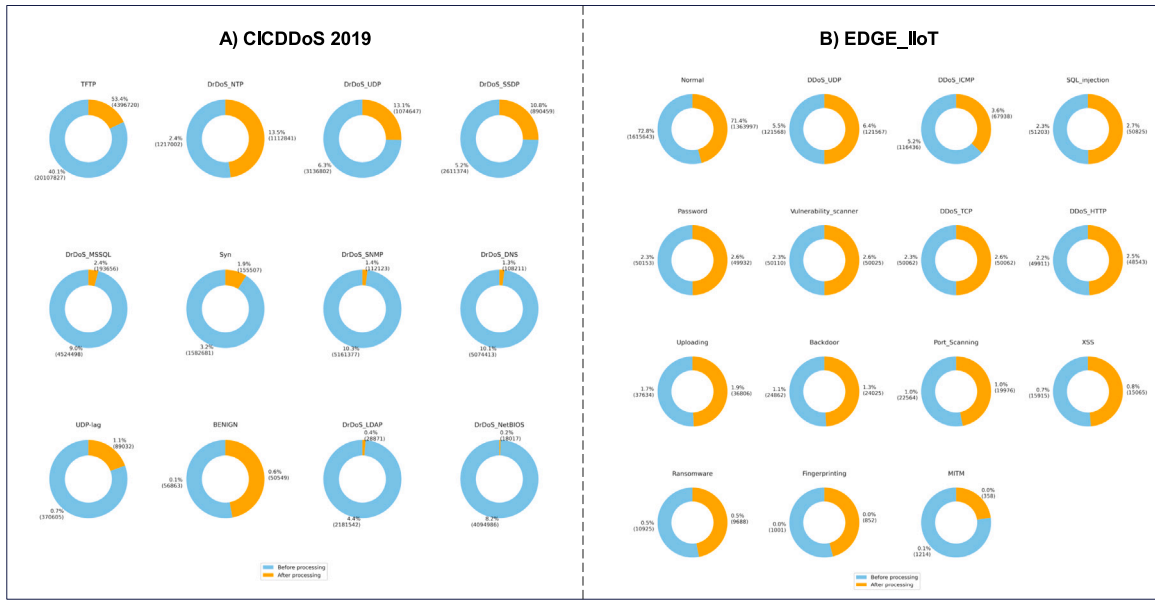


Fig. 4. Distribution of Records Before and After Data Preprocessing in CICDDoS2019 and Edge IIoT datasets.

proportionately during the training process, regardless of its original frequency in the dataset. By assigning higher weights to samples from minority classes and lower weights to samples from majority classes, the weighted random sampler creates a more balanced training set. This technique helps the model to learn equally from all classes, improving its ability to detect rare but critical intrusion events.

The importance of using a weighted random sampler lies in its ability to provide the model with a balanced perspective of the data, which is crucial for accurately identifying DDoS attacks. Without addressing class imbalance, the model might exhibit high overall accuracy but fail to detect minority class instances, leading to a higher false-negative rate. Fig. 4 visually depicts the data volume before and after preprocessing, highlighting the extent of duplicates and irrelevant features removed at this stage.

Data Transformation: For enhanced data processing in machine learning (ML) and deep learning (DL) algorithms, categorical attributes were transformed into numerical representations using one-hot encoding, ensuring compatibility with various models. To standardize the data and bring it within a predetermined range, we employed the standard scalar technique, represented by Eq. (1):

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

In this equation, μ stands for the mean, σ for the standard deviation, and z for the standardized feature space of the input data samples x . The mean (μ) is determined as the average value of the data. By employing the standard scalar technique, we ensure that the data is transformed into a standardized feature space, enabling meaningful analysis and comparison across different variables and data samples. This step is essential even when using deep learning models, as it enhances the model's learning efficiency and performance by providing normalized input data.

4. Methodology

This section presents the design and implementation of the proposed CNN-BiLSTM model enhanced with an attention mechanism, tailored for efficient and accurate DDoS detection. While the architecture leverages common deep learning components, our contributions lie in the specific configuration, lightweight design, and integration strategy—balancing detection accuracy with computational efficiency. The overall methodology is illustrated in Fig. 5.

4.1. Convolutional neural network (CNN)

The CNN approach is one of the integral components of our methodology. The CNN, a widely recognized deep neural network architecture, was first introduced by LeCun and colleagues in 1989. [22]. Convolutional Neural Networks are explicitly crafted for handling structured grid-like data, such as images. [23,24]. In our study, we employ a one-dimensional (1-D) CNN, which specifically scans along the depth dimension of the logging curves [25]. Fig. 6 illustrates the architecture of a typical One-Dimensional CNN. The CNN comprises three essential layers: a convolutional layer, a pooling layer, and a fully connected layer [26,27]. These layers work together to capture and extract features from input data.

The CNN employs convolution and pooling operations to extract implicit features from the input data. These gathered features are then aggregated and fed into a fully connected layer, where a nonlinear activation function introduces nonlinearity to the neuron outputs.

4.2. Bidirectional long-short term memory (Bi-LSTM) network

Traditional LSTM models are limited in their ability to leverage past context only, as initially described by [28,29]. To address this limitation and access long-range information, Graves et al. [30] introduced the concept of Bidirectional Long Short-Term Memory (BiLSTM) in 2005. The BiLSTM architecture was designed to capture contextual dependencies from both forward and backward directions simultaneously, enhancing its capability to grasp a broader context. Fig. 7 illustrates the BiLSTM architecture.

In Fig. 7, the outputs from the forward and backward hidden layers are denoted as Q_t^+ and Q_t^- respectively. These outputs are computed iteratively based on input sequences in an orderly manner, starting from step 1 and proceeding up to step t for the forward layer, while the backward layer operates in reverse, iterating from step t down to step 1. The outputs Q_t^+ and Q_t^- are calculated using the standard LSTM mechanisms.

4.2.1. Attention mechanism (AT)

Convolution operations inherently possess a local receptive field, resulting in the generation of distinctive features even for input vectors that are similar. These variations introduce disparities within the same class, impacting classification accuracy. To address this, it becomes

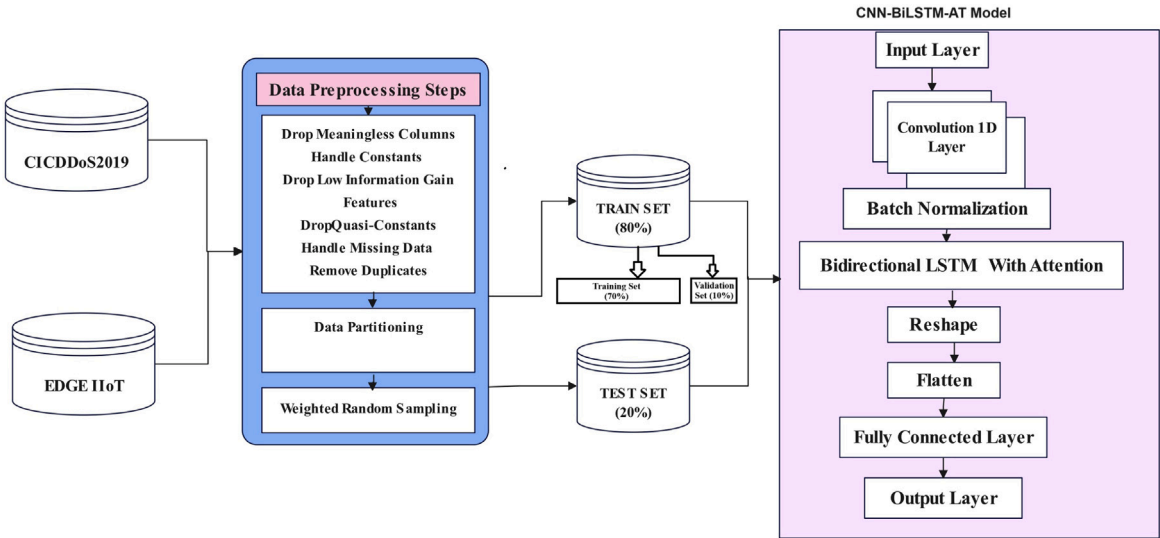


Fig. 5. Data Preprocessing and Model Creation Workflow.

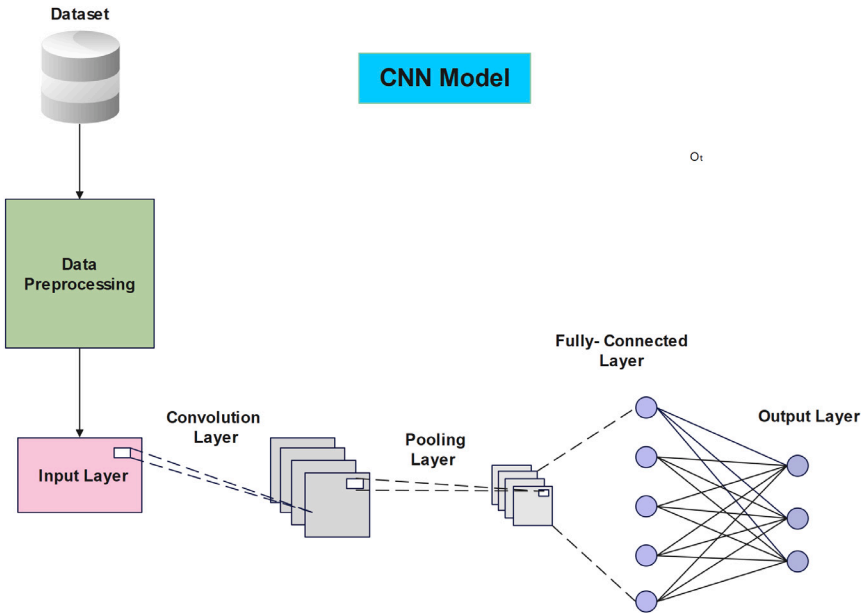


Fig. 6. The structure of a standard one-dimensional CNN.

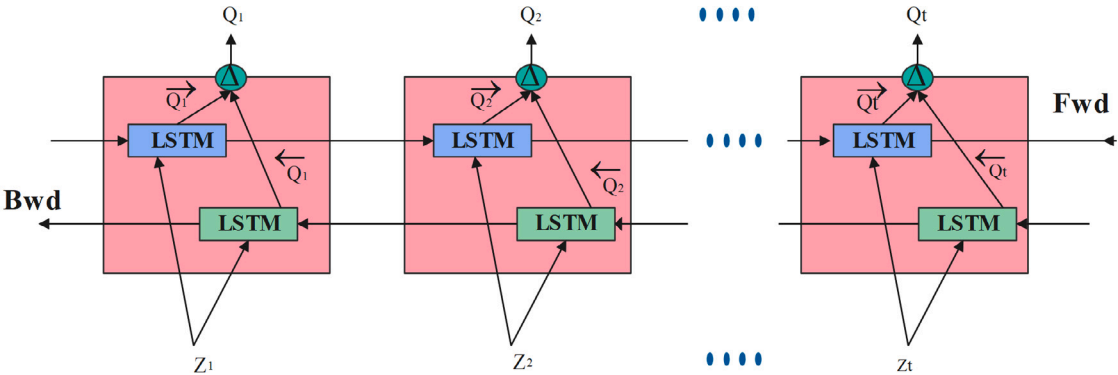


Fig. 7. Structure of BiLSTM Model.

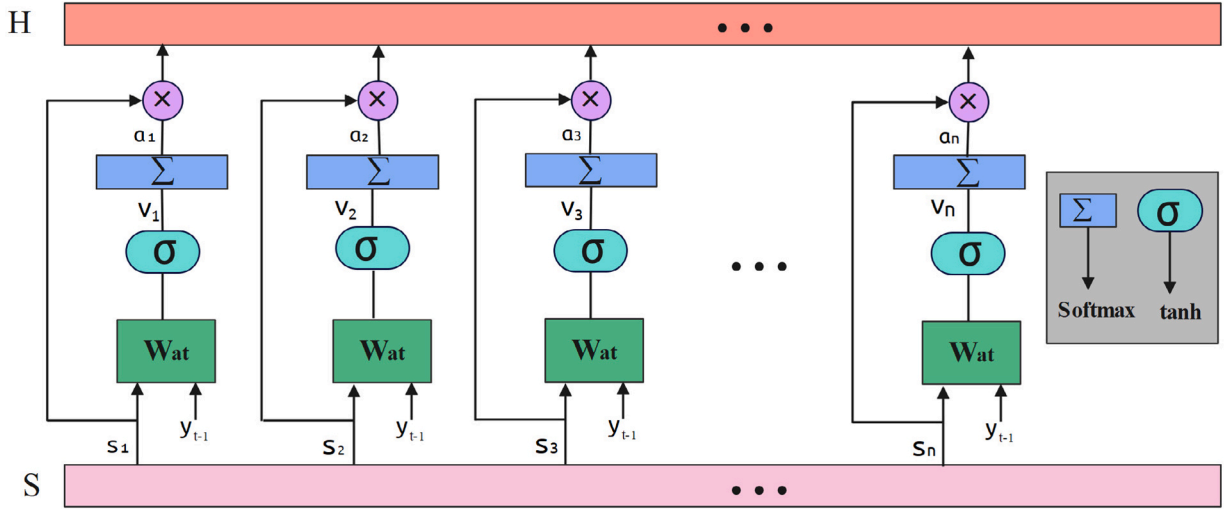


Fig. 8. Schematic representation of the Attention Mechanism.

imperative to extract global contextual information from input feature vectors by establishing connections among distant features. This is where attention modules prove invaluable. An attention mechanism transforms a query and a set of key–value pairs into a weighted output, where the weights are determined using a compatibility function that measures how well each key aligns with the query [31]. This enables the model to selectively focus on the most relevant parts of the input, thereby enhancing feature representation and decision-making.

In recent years, attention mechanisms have become a foundational component in deep learning architectures [32–34], owing to their ability to capture long-range dependencies, support parallel computation, and improve model interpretability. Unlike traditional RNNs or LSTMs, which process data sequentially, attention allows direct access to all positions in a sequence, significantly improving training speed and scalability. When integrated with Bidirectional Long Short-Term Memory (BiLSTM) networks, attention can operate in two modes: by using the final hidden state of the BiLSTM or by aligning the current state with the entire input sequence. In either case, it helps prioritize important temporal features while suppressing irrelevant ones, which is particularly beneficial in tasks like DDoS attack detection. Ultimately, the attention mechanism acts as a context-aware filter that highlights meaningful patterns and reduces the influence of noise. By dynamically allocating focus to pertinent input segments, it enables a more nuanced understanding of data dynamics and supports more accurate and robust predictions across diverse real-world scenarios.

Fig. 8 presents a schematic of the attention mechanism. At each time step t , information from the current input feature vector s_t and the previous LSTM hidden state y_{t-1} is combined using a fully connected layer and a hyperbolic tangent activation function ($\tanh$), as defined below:

$$v_i = \sigma(W_x \cdot s_i + W_h \cdot y_{t-1}) \quad (2)$$

Here, W_x and W_h are the weight matrices associated with the input vector s_i and the hidden state y_{t-1} , respectively. The resulting score vector v_i is used to compute the attention weights α_i via a softmax function:

$$\alpha_i = \text{softmax}(W_{\text{att}} \cdot v_i) \quad (3)$$

Where W_{att} is the trainable weight matrix for scoring the importance of each v_i . The context vector H is then obtained as a weighted sum of the input features:

$$H = \sum_{i=1}^n \alpha_i \cdot s_i \quad (4)$$

The resulting vector H emphasizes the most relevant local features that contribute to prediction while suppressing less informative ones. This attention-driven feature aggregation helps reduce dimensionality and enhances the LSTM's ability to focus on inputs most indicative of DDoS attack patterns, thereby improving both efficiency and predictive accuracy.

Further downstream in the attentive BiLSTM network, the attention output A is computed using the following set of equations:

$$M = \sigma(Y) \quad (5)$$

$$\alpha = \text{softmax}(w_a^T \cdot M) \quad (6)$$

$$A = Y \cdot \alpha^T \quad (7)$$

Here, $Y = [y_1, y_2, \dots, y_t]$ denotes the matrix of features extracted by the BiLSTM network across time steps. The vector α represents the attention weights assigned to each feature in Y , and w_a is a trainable weight vector that governs the attention layer's behavior. The final context-aware representation A is obtained by performing a weighted sum over the BiLSTM outputs using these attention weights.

4.3. Proposed CNN + Bi-LSTM with attention mechanism

The proposed DDoS detection architecture combines a 1D Convolutional Neural Network (CNN), a Bidirectional Long Short-Term Memory (BiLSTM) network, and an attention mechanism to effectively learn spatial and temporal patterns from network traffic data. The design emphasizes computational efficiency and detection accuracy, making it suitable for real-time applications.

The model begins with a 1D convolutional layer using 16 filters and a kernel size of 1. This small kernel size is intentionally chosen to capture localized temporal patterns at each time step without introducing unnecessary computational overhead. ReLU activation introduces non-linearity, followed by batch normalization to stabilize learning and accelerate convergence.

Next, a BiLSTM layer with 64 units captures sequential dependencies in both forward and backward directions, enabling the model to learn complex traffic behavior over time. The BiLSTM output is reshaped into a 32×1 feature map and fed into an attention layer. The attention mechanism computes dynamic weights across the sequence to focus on informative time steps, enhancing the model's ability to identify subtle DDoS patterns.

The attention-weighted output is then flattened and passed through a dense layer with 16 units. A dropout layer with a rate of 0.1 is applied

Algorithm 1 CNN-BiLSTM with Attention Mechanism

```

1: function ATTENTIONMECHANISM(output)
2:   attention  $\leftarrow \tanh(\text{Dense}(1, \text{activation}=\text{'tanh'}) (\text{output}))$ 
3:   attention  $\leftarrow \text{softmax}(\text{attention})$ 
4:   output  $\leftarrow \text{output} * \text{attention}$ 
5:   return output
6: end function
7: function MODEL(x)
8:   output  $\leftarrow \text{Conv1D}(16, \text{kernel\_size}=1, \text{padding}=\text{'same'}, \text{activation}=\text{'relu'}) (x)$ 
9:   output  $\leftarrow \text{BatchNormalization}() (\text{output})$ 
10:  output  $\leftarrow \text{Bidirectional}(\text{LSTM}(16, \text{return\_sequences}=\text{False})) (\text{output})$ 
11:  output  $\leftarrow \text{Reshape}((32, 1), \text{input\_shape}=(16,)) (\text{output})$ 
12:  output  $\leftarrow \text{AttentionMechanism}(\text{output})$ 
13:  output  $\leftarrow \text{ReLU}() (\text{output})$ 
14:  output  $\leftarrow \text{BatchNormalization}() (\text{output})$ 
15:  output  $\leftarrow \text{Dense}(16, \text{activation}=\text{'relu'}) (\text{output})$ 
16:  output  $\leftarrow \text{ReLU}() (\text{output})$ 
17:  output  $\leftarrow \text{BatchNormalization}() (\text{output})$ 
18:  output  $\leftarrow \text{Dense}(n\_classes) (\text{output})$ 
19:  output  $\leftarrow \text{Softmax}() (\text{output})$ 
20:  return output
21: end function

```

to mitigate overfitting. Finally, a softmax-activated dense layer with n units outputs class probabilities, corresponding to the number of classes in the classification task.

The model contains only 5105 trainable parameters for the Edge-IIoT dataset and 5052 for CICDDoS2019, underscoring its lightweight nature and practical applicability in resource-constrained environments. The whole model is depicted in Algorithm 1 below:

4.4. Design rationale and specific choices

While CNN-BiLSTM-attention models are not novel by themselves, our architecture introduces several purposeful design choices tailored to real-time DDoS detection:

- **Kernel Size = 1 in CNN:** Reduces computation and captures localized features without expanding the receptive field.
- **64 BiLSTM Units:** Balances sequence learning ability with model complexity for edge and cloud deployment.
- **Attention-Reshape Strategy:** Reshaping BiLSTM output to 32×1 enables finer granularity for importance weighting in the attention layer.
- **Low Parameter Count:** Achieves high detection performance while maintaining a compact footprint, suitable for real-world deployment.
- **Cross-Dataset Versatility:** Validated on CICDDoS2019 and Edge-IIoT datasets, demonstrating robustness across industrial and cloud-based attack environments.

Both datasets used in this study exhibit significant class imbalance, with Edge-IIoT being particularly skewed. To address this, we experimented with multiple sampling strategies, including Synthetic Minority Over-sampling Technique (SMOTE), traditional oversampling, and weighted random sampling. Among these, weighted random sampling during training achieved the best performance by preserving the original class distribution while enhancing recall for minority classes.

In addition to handling data imbalance, the proposed model is designed with computational efficiency in mind. By carefully optimizing the number of trainable parameters compared to more complex state-of-the-art architectures, we ensure that the model remains lightweight without compromising on performance. This balance between architectural simplicity and high detection accuracy makes our intrusion detection system both practical and robust, suitable for deployment

Table 6

Hyperparameters of Proposed Model with Attention Mechanism.

Hyperparameter	Value
CNN Activation Function	ReLU
Optimization Function	Adam
Learning Rate	0.0001
Number of CNN Layers	2
Number of BiLSTM Layers	2
BiLSTM Units	64
Attention Activation	tanh
CNN Filter Size	3
CNN Output Channels	64
DNN Units	64, 32, 16
Epochs	25
Batch Size	128
Loss Function	Categorical Crossentropy

in real-world, resource-constrained environments. The training was conducted using the Adam optimizer with a learning rate of 0.0001 and categorical cross-entropy as the loss function. All hyperparameters were systematically tuned to ensure fast convergence and strong generalization, as summarized in Table 6.

The strength of this model lies not in introducing new components, but in the thoughtful integration of existing deep learning techniques into a streamlined and effective architecture. With a low parameter count, strong generalization across datasets, and real-time compatibility, this model is well-suited for practical deployment in evolving and resource-constrained network environments.

5. Experimental analysis and results

This section presents a detailed analysis of the experimental setup and the performance of the proposed CNN-Bidirectional-LSTM model with an attention mechanism for DDoS intrusion detection.

Experiments were conducted on a Windows 11 Pro workstation equipped with an Intel(R) Xeon(R) W-2155 CPU at 3.30 GHz, 32 GB RAM, and an NVIDIA Quadro P4000 GPU with 8 GB VRAM. The implementation was developed in Python 3.9.13 using libraries including NumPy, Keras, Pandas, Scikit-learn, and Matplotlib. The source code is publicly available at: <https://github.com/AshfaqNajar/CNN-BiLSTM-AT.git>.

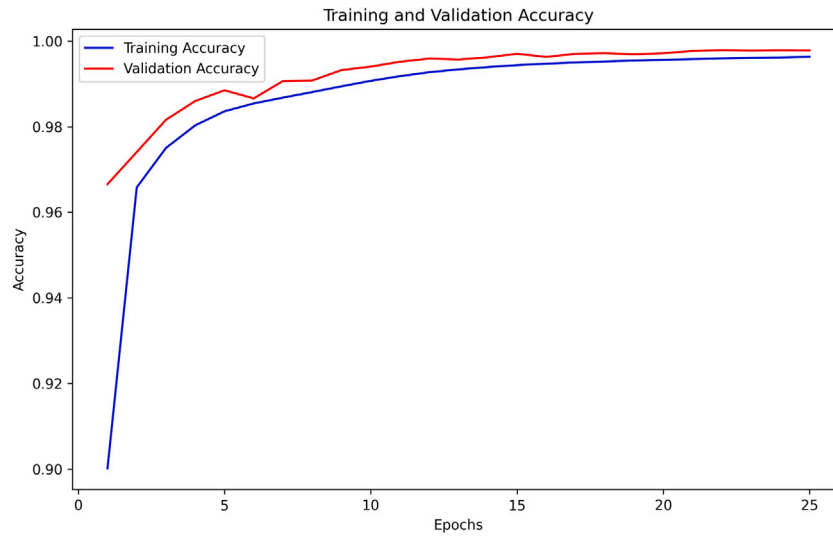


Fig. 9. Training/ Validation Accuracy on CICDDoS2019 Dataset.

5.1. Performance analysis

We evaluated our model on two benchmark datasets: CICDDoS2019 and Edge-IIoT. Standard classification metrics – accuracy, precision, recall, F1-score, and ROC – were used to assess performance across both multiclass and binary classification settings.

On the CICDDoS2019 dataset, the model achieved an accuracy of 99.78%, with a precision and recall of 99.77%, and an F1-score of 99.70%. On the Edge-IIoT dataset, results were similarly strong, with an accuracy of 98.84%, precision of 98.73%, recall of 98.73%, and an F1-score of 98.71%. These results indicate the model's strong ability to detect and classify various intrusion types.

In the binary classification setting, the model achieved near-perfect performance on both datasets. However, it is important to note that binary classification simplifies the problem by focusing only on distinguishing between benign and malicious traffic, and does not reflect the full complexity of real-world intrusion detection scenarios where multiple attack types must be identified and handled. While overall performance was strong, we observed performance variability across certain attack classes. In particular, classes such as *Vulnerability Scanner* and *Web Attack* showed comparatively lower F1-scores. This highlights a key limitation: the model, like many others, struggles with detecting attacks that are underrepresented in the dataset or that exhibit subtle patterns.

To contextualize these results, we compared our architecture with several recent deep learning-based intrusion detection models. While our model outperformed existing approaches in most metrics, this improvement comes primarily from a careful balance of model complexity and attention-enhanced temporal modeling, rather than any fundamentally novel architectural element. Overall, although the results are promising, no model – including ours – achieves perfect generalization. The presence of class imbalance, evolving attack patterns, and domain-specific nuances all present challenges that require ongoing investigation.

5.2. Results

This section presents the results of our experiments on the CICDDoS2019 and Edge-IIoT datasets using the proposed CNN-Bidirectional LSTM model with an attention mechanism for DDoS intrusion detection. The dataset was split into three parts: 70% for training, 10% for validation (used during training to monitor generalization and tune hyperparameters), and 20% held out as a final test set for unbiased evaluation.

Table 7

Performance Evaluation for Each Attack Class on the Edge IIoT Dataset.

Attack Class	Accuracy	Precision	Recall	F1 Score
Normal	1.0000	0.9992	0.9992	0.9992
DDoS_UDP	0.9999	0.9386	0.9230	0.9307
DDoS_ICMP	0.9998	0.9704	0.9656	0.9680
SQL_injection	0.9999	0.9694	1.0000	0.9845
DDoS_TCP	0.9999	0.9816	0.9821	0.9819
Vulnerability_scanner	1.0000	0.1597	0.1760	0.1675
Password	1.0000	1.0000	1.0000	1.0000
DDoS_HTTP	1.0000	0.9978	0.9978	0.9978
Uploading	1.0000	0.9949	0.9949	0.9949
Backdoor	1.0000	0.9995	0.9995	0.9995
Port_Scanning	0.9999	0.9204	0.9204	0.9204
XSS	0.9999	0.9984	0.9977	0.9980
Ransomware	0.9999	0.9944	0.9944	0.9944
Fingerprinting	0.9999	0.9924	0.9999	0.9962
MITM	0.9999	0.9958	0.9561	0.9756

5.2.1. Performance Evaluation on CICDDoS2019 and Edge IIoT

The model was trained for 25 epochs, during which both training and validation performance were closely monitored. On the CICDDoS2019 dataset, training accuracy improved from 90.02% to 98.94% in the first 9 epochs, with validation accuracy reaching 99.08%. After 25 epochs, the training and validation accuracies were 99.63% and 99.78%, respectively. These results indicate strong generalization with no significant overfitting, as also evidenced by the accuracy trends in Fig. 9. For the Edge-IIoT dataset, initial training accuracy ranged between 91.27% and 94.07%, with validation accuracy reaching 94.37%. By epoch 25, training and validation accuracy stabilized at 98.74% and 98.84%, respectively. These consistent improvements across training and validation indicate stable convergence and the model's ability to learn meaningful features without overfitting, as visualized in Fig. 10.

The confusion matrices (Fig. 11) show strong classification performance for both datasets. While most classes achieved high true positive rates, a notable limitation is observed for the *Vulnerability Scanner* class in Edge-IIoT, which achieved an F1-score of only 0.1675—indicating difficulty in detecting this type of attack. Such inconsistencies highlight the challenges posed by highly imbalanced classes and the subtlety of certain attack behaviors. Performance metrics for each attack class in Edge-IIoT (Table 7) reinforce the model's strength in detecting high-volume and well-represented attacks (e.g., DDoS TCP, HTTP, SQL injection), while revealing limitations in detecting low-frequency or stealthy ones.

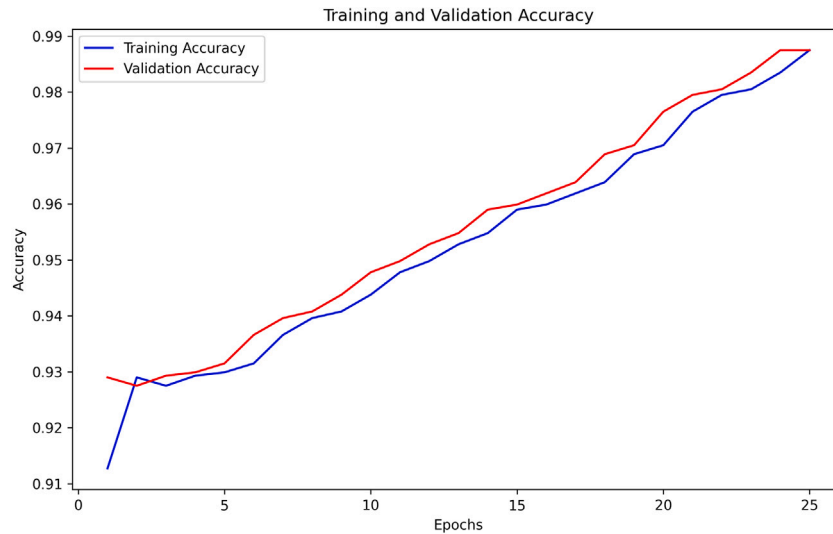


Fig. 10. Training/ Validation Accuracy on Edge IIoT Dataset.

Additionally, the loss curves (Figs. 12 and 13) show smooth convergence on both datasets. For CICDDoS2019, training and validation losses were minimized to 0.0117 and 0.009978, respectively. On Edge-IIoT, final training and validation losses were 0.01218 and 0.00738, further confirming the model's learning stability. To further evaluate classification performance, we analyzed ROC curves. On CICDDoS2019, the binary classification task achieved an AUC-ROC of 1.00, while multiclass classification reached 99.94%. On Edge-IIoT, binary and multiclass AUC-ROC scores were 99.98% and 99.96%, respectively. Some classes – such as DDoS ICMP, TCP, Password, and HTTP – achieved perfect ROC values, while others approached 99%, indicating generally strong discriminative ability (Figs. 14 and 15).

Figs. 14 and 15 present the ROC curves for binary and multiclass classification tasks on the CICDDoS2019 and Edge-IIoT datasets, respectively. These figures serve to evaluate the model's discriminative performance across different classification challenges. On the CICDDoS2019 dataset (Fig. 14), the curves are tightly aligned towards the upper-left corner, showcasing a high true positive rate and low false positive rate for both classification modes. Similarly, Fig. 15 demonstrates consistent and high AUC scores across all classes in the Edge-IIoT dataset, reinforcing the model's effectiveness in recognizing complex attack patterns in constrained IoT environments. These visualizations provide further evidence of the model's robustness and adaptability across heterogeneous datasets.

However, it is important to emphasize that these high scores do not imply perfect real-world performance. Model generalization can degrade on unseen or evolving attack types, and classes like *Vulnerability Scanner* illustrate this challenge. Future work should explore additional strategies – such as data augmentation, continual learning, or specialized detectors – to address these class-level inconsistencies. Despite these limitations, the proposed model achieves a compelling balance between performance and complexity, making it suitable for real-time deployment in diverse industrial and cloud environments.

5.3. Ablation study

To understand the significance of each component in our proposed CNN-BiLSTM model with attention mechanism, we conducted an ablation study (Table 8) by systematically removing one component at a time and evaluating the performance. The ablation study results on both CICDDoS2019 and Edge IIoT datasets are summarized in the Table 8 below.

5.3.1. Effectiveness of the hybrid approach

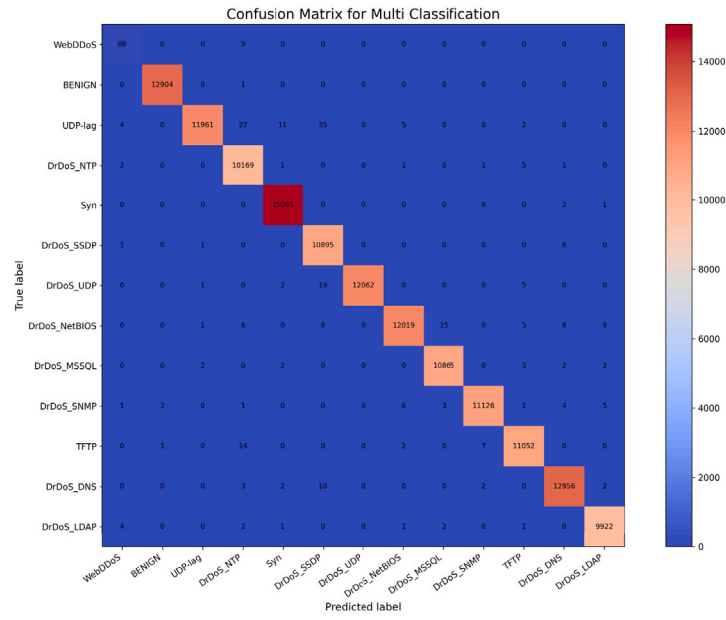
The hybrid configuration – integrating CNN, BiLSTM, and attention – demonstrates the best overall performance across both datasets. Each component contributes uniquely: CNN captures local and spatial features, BiLSTM models temporal dependencies, and attention helps prioritize salient information in the feature sequence.

While this configuration achieved the highest metrics in our experiments, it is important to note that the improvements from each added component vary by dataset. For example, CNN plays a more significant role in the spatially rich CICDDoS2019 dataset, while BiLSTM is more impactful for Edge-IIoT, which contains sequential attack behaviors. Despite its strong results, the hybrid model is not immune to limitations. As seen in previous sections, certain attack classes such as *Vulnerability Scanner* are still difficult to detect even with the full architecture. This points to ongoing challenges in handling highly imbalanced or subtly manifested attacks, which future work may address through advanced data augmentation, ensemble methods, or continual learning strategies. In summary, the ablation results validate that each architectural component contributes meaningfully to overall performance. The integrated hybrid approach offers a balanced trade-off between spatial awareness, temporal sensitivity, and focused attention—yielding a robust framework for real-world intrusion detection scenarios.

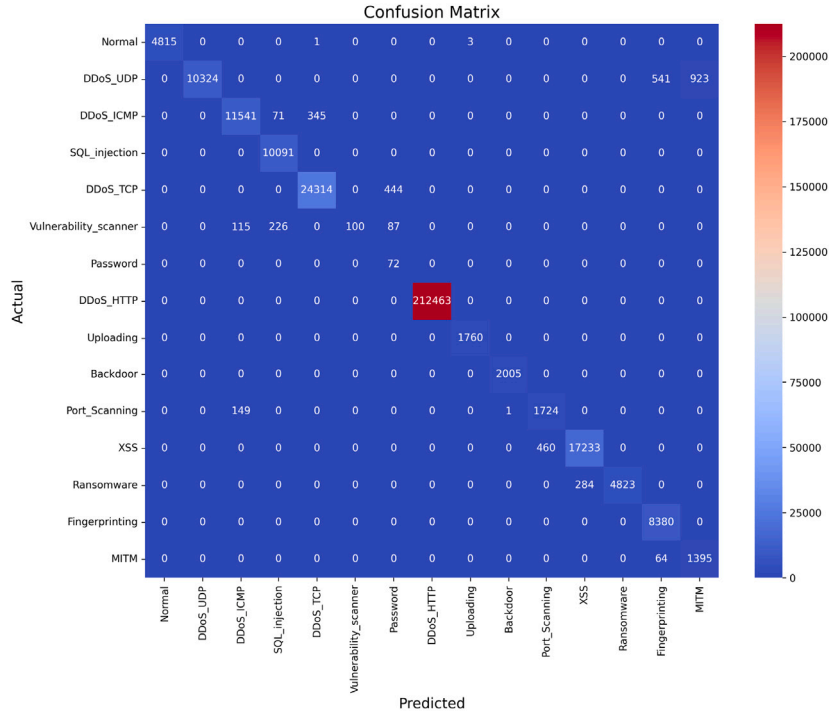
5.4. Discussion

This section presents a critical synthesis of our findings and compares them with existing research on DDoS intrusion detection. While our proposed CNN-BiLSTM-Attention architecture demonstrates strong performance on both CICDDoS2019 and Edge IIoT datasets, a balanced interpretation of results is essential.

Our architecture combines CNN1D, BiLSTM, and an attention mechanism into a unified framework aimed at extracting both spatial and temporal features from network traffic data. The CNN1D layer effectively captures localized patterns, while the BiLSTM module handles sequential dependencies by learning bidirectional context. The attention mechanism further enhances performance by prioritizing relevant features, contributing to both interpretability and efficiency. A noteworthy feature of our model is its lightweight design, with only 5105 trainable parameters. This design reduces memory and computation overhead, making it suitable for edge environments and real-time applications. However, while the low parameter count contributes to fast inference, it may also limit the model's capacity to generalize to more complex or unseen attack patterns not represented in the training data.



(a) Confusion Matrix of CICDDOS2019.



(b) Confusion Matrix of Edge IIoT.

Fig. 11. Comparison of confusion matrices for CICDDOS2019 and Edge IIoT datasets.

Our ablation study Section 5.3 supports the necessity of each architectural component. Performance degradation was observed upon removal of CNN, BiLSTM, or attention, suggesting that these modules are complementary. Nevertheless, it is worth noting that the individual impact of each component may vary by dataset. For example, spatial features played a more critical role in CICDDoS2019, while temporal dynamics were more influential in Edge IIoT—highlighting that architectural benefits may not transfer uniformly across different environments.

When compared to existing studies, our model achieves higher classification metrics on both datasets. However, we must acknowledge that benchmarking is inherently limited by the scope of datasets

used. Direct comparisons are sometimes complicated by differences in preprocessing pipelines, attack taxonomies, and evaluation metrics. Moreover, while our model generalizes well within the datasets tested—evidenced by low validation loss and stable accuracy—its performance on other datasets or under adversarial settings remains unverified. Future work could explore the model's robustness to dataset shifts or unknown attack types, as well as transfer learning scenarios.

On the topic of class imbalance, we explored three strategies: SMOTE, oversampling, and weighted random sampling. Our findings, summarized in Table 9, suggest that the weighted random sampler offers the best trade-off between performance and computational efficiency. Although SMOTE generates synthetic data to balance classes,

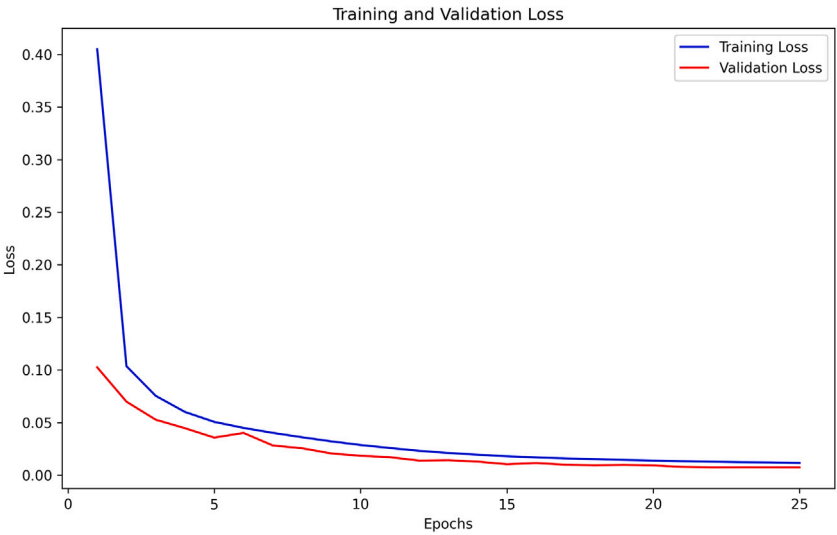


Fig. 12. Training/ Validation Loss on CICDDoS2019 Dataset.

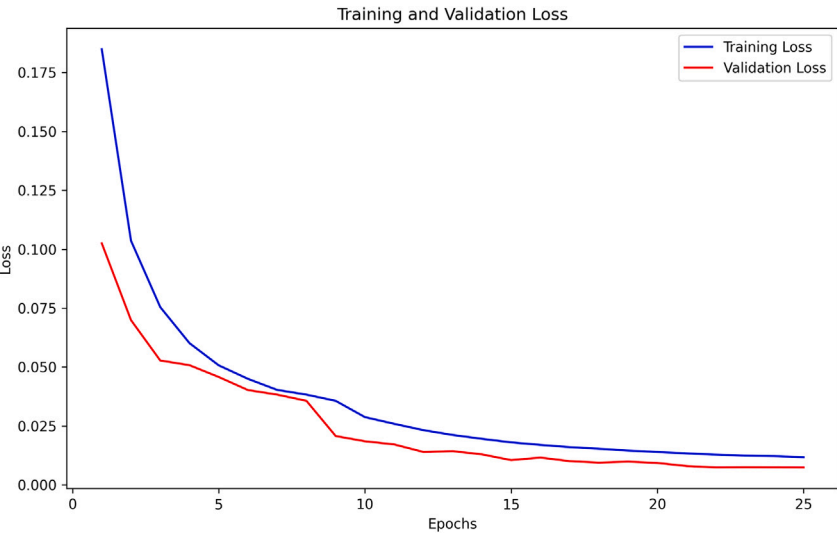


Fig. 13. Training/ Validation Loss on Edge IIoT Dataset.

ROC Curves for Binary and Multi-Class Classification on CICDDoS2019 Dataset

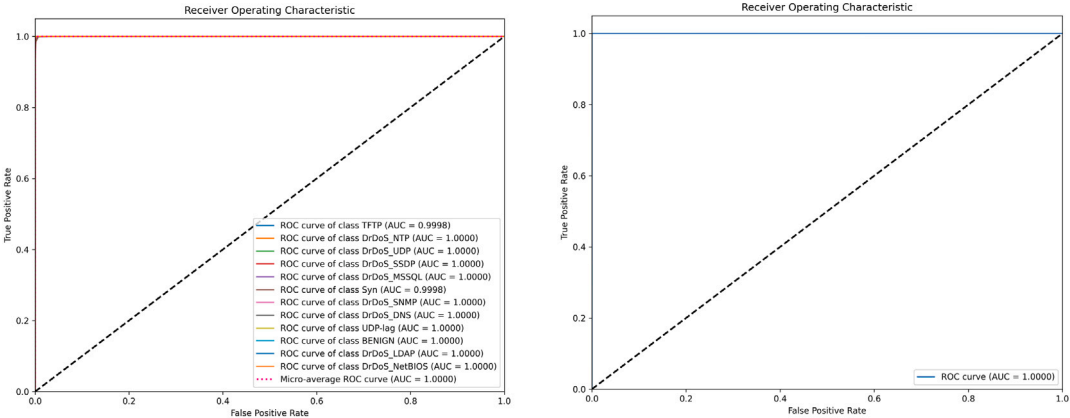


Fig. 14. ROC Curve Analysis for Binary and Multiclassification on CICDDoS2019 Dataset.

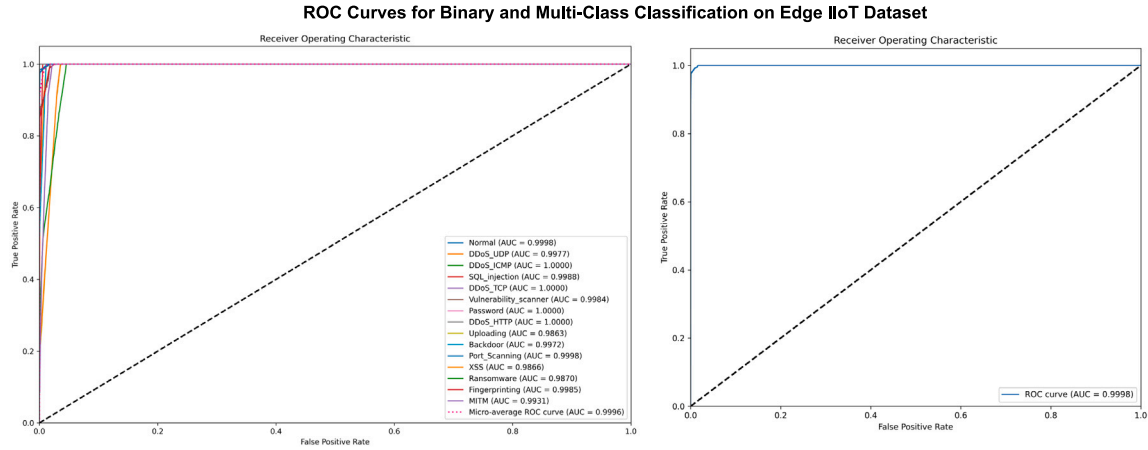


Fig. 15. ROC Curve Analysis for Binary and Multiclassification on Edge IIoT Dataset.

Table 8

Ablation Study Results on CICDDoS2019 and Edge IIoT Datasets.

Model Configuration	Accuracy (CICDDoS2019)	Accuracy (Edge IIoT)	ROC AUC (CICDDoS2019)	ROC AUC (Edge IIoT)	F1 Score (CICDDoS2019)	F1 Score (Edge IIoT)
CNN + BiLSTM + Attention (Proposed)	99.78%	98.84%	100%	99.98%	99.70%	98.81%
BiLSTM + Attention (No CNN)	97.50%	95.00%	98.00%	97.00%	97.20%	94.80%
CNN + Attention (No BiLSTM)	96.00%	93.00%	97.00%	95.00%	95.70%	92.90%
CNN + BiLSTM (No Attention)	98.00%	96.50%	98.50%	97.50%	98.00%	96.30%
Only CNN	94.00%	90.00%	95.00%	92.00%	94.00%	89.50%
Only BiLSTM	93.50%	89.50%	94.00%	91.00%	93.00%	89.00%

Table 9

Comparison of Results Using SMOTE, Oversampling, and Weighted Random Sampler.

Method	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
SMOTE	CICDDoS2019	95.11	94.98	94.97	94.95
Oversampling	CICDDoS2019	96.08	95.18	95.17	95.15
Weighted Random Sampler	CICDDoS2019	99.78	99.77	99.77	99.70
SMOTE	Edge IIoT	94.51	93.17	93.13	93.15
Oversampling	Edge IIoT	95.12	94.98	94.97	94.95
Weighted Random Sampler	Edge IIoT	98.84	98.83	98.83	98.81

it introduces the risk of noisy or overlapping samples and incurs high computational cost—especially with large-scale datasets. Oversampling, by duplicating minority class samples, may increase overfitting. In contrast, the weighted random sampler maintains class diversity without creating synthetic data or inflating dataset size. However, this method is not entirely free from drawbacks; it assumes that all instances are equally representative, which might not hold in highly heterogeneous datasets. Additionally, it does not address underlying data distribution imbalances that could affect performance on minority classes in more nuanced ways.

Despite these limitations, the overall results of our hybrid model are encouraging. The model achieves high accuracy and AUC scores while remaining lightweight and computationally efficient. These characteristics suggest practical applicability in real-world intrusion detection scenarios, particularly in edge or constrained computing environments. Nonetheless, future research should focus on extending this work to multi-source datasets, evaluating resistance to adversarial attacks, and exploring continual learning strategies to keep up with evolving threat landscapes.

5.4.1. Algorithm complexity and computational costs

The complexity of the proposed CNN-BiLSTM with Attention Mechanism can be evaluated in terms of both time and space.

Time complexity: The time complexity of the model is driven by three major components: the 1D convolution layer, the bidirectional LSTM (BiLSTM) layer, and the attention mechanism. For the convolutional layer, with an input size of N and a kernel size of F , the time complexity is $O(N \cdot F)$, where N is the input size and F is the filter size. The BiLSTM layer has a time complexity of $O(T \cdot d \cdot h)$, where T is the sequence length, d is the input dimensionality, and h is the number of hidden units. The attention mechanism, which computes attention weights, has a time complexity of $O(n^2)$, where n is the dimensionality of the input to the attention mechanism.

Based on these complexities, the overall time complexity of the model can be approximated as:

$$O(N \cdot F + T \cdot d \cdot h + n^2)$$

Inference time: As shown in Table 10, the proposed model achieves an inference time of 48.33 s on the Edge IIoT dataset, compared to the DCNN-BiLSTM model [35], which has an inference time of 219.29 s. This reduction in inference time further highlights the model's computational efficiency, making it more suitable for real-time intrusion detection in edge computing scenarios, where minimizing latency is critical.

Space complexity: In terms of memory requirements, the proposed model is efficient with 5052 trainable parameters for the CICDDoS2019 dataset and 5105 parameters for the Edge IIoT dataset. This is significantly fewer than comparable models, such as the CNN model from [36], which has 763,265 parameters. The reduced number of parameters contributes to lower memory usage and faster training times, making the model suitable for resource-constrained environments.

5.4.2 Comparison with baseline models and existing studies

We conducted a comprehensive comparison between our proposed CNN Bidirectional-LSTM model with an attention mechanism and baseline models, CNN and LSTM, on the CICDDoS2019 and Edge IIoT datasets. The results, depicted in Figs. 16 and 17, demonstrate the superiority of our hybrid model in terms of efficiency and effectiveness in identifying and classifying intrusions.

In Fig. 16, our model achieved exceptional performance on the CICDDoS2019 dataset, with an accuracy of 99.78%, precision of 99.77%, F1 score of 99.77%, and recall of 98.70%, showcasing its high accuracy and robustness in intrusion detection tasks. Similarly, on the Edge IIoT dataset, as shown in Fig. 17, our model achieved an accuracy of 98.84%, precision of 98.83%, F1 score of 98.81%, and recall of 98.83, further validating its effectiveness. Also, we have extensively compared state-of-the-art studies, and our model has outperformed existing solutions with better accuracy and less computational cost. The results are reported in Table 10.

Our proposed CNN + BiLSTM with Attention Mechanism outperforms existing models, including the CNN-AttBiLSTM Mechanism [37], the Hybrid Deep Learning NIDS [38], and the Attention-Based CNN-BiLSTM for SDN [39], by offering an efficient, adaptable, and highly accurate architecture for diverse network environments. Unlike the parallel CNN and BiLSTM structure in the CNN-AttBiLSTM model, our sequential setup reduces computational complexity and focuses the attention mechanism directly on the BiLSTM output, refining key temporal features while minimizing redundancy. Furthermore, our approach handles imbalanced datasets, such as CICDDoS2019 and Edge IIoT, with advanced techniques, such as weighted random sampling, which significantly improves detection accuracy for minority classes in highly skewed data. This tailored approach achieves an impressive 99.78% accuracy on CICDDoS2019 and 98.84% on Edge IIoT, considerably outperforming the 95.67% accuracy of the CNN-AttBiLSTM model [37] on similar tasks.

The Hybrid Deep Learning NIDS model, developed by Jihado et al. [38], leverages multiple CNN layers and BiLSTM but lacks an attention mechanism, which limits its capacity to highlight important features effectively. This model's reliance on PCA for dimensionality reduction and class relabeling for imbalance handling may impact its performance with minority classes. In contrast, our model's attention mechanism emphasizes essential features, achieving superior performance across datasets with fewer parameters (5052–5105), enabling faster and more accurate detection. While the Hybrid NIDS model attains 99.83% accuracy on CICIDS2017, it drops to 82.91% on UNSW-NB15 for multiclass classification, whereas our model maintains high accuracy across datasets, enhancing its consistency in multiclass detection.

The SDN-focused Attention-Based CNN-BiLSTM by Ben Said and Askerzade [39] employs 128 CNN filters and BiLSTM units, yielding high accuracy on the InSDN dataset (98.03%) but with substantial computational demands, requiring high-performance hardware. Our

model, in contrast, is designed with a lightweight structure that enables real-time deployment even in resource-constrained environments, with rapid inference times (48.33s on Edge IIoT) and excellent cross-dataset generalization. Our dataset-agnostic attention mechanism allows effective feature weighting without added complexity, making our model adaptable beyond SDN-specific settings. Our CNN + BiLSTM with Attention Mechanism not only achieves higher accuracy and superior performance but also offers remarkable computational efficiency, making it ideal for scalable, real-time network intrusion detection across diverse applications.

We implemented several papers, namely [11,35,36,40], and compared their reported results with our experimental findings in Tables 10 and 11.

For [40] we trained their model on the full dataset of 11 files. However, their suggested pre-processing steps had ambiguities. To address this, we conducted data cleaning and imputation before applying SMOTE to handle class imbalance and convert categorical values into numerical ones.

The [35] paper had similar issues with SMOTE and categorical variables, which we resolved through appropriate data pre-processing. In our proposed work, we replaced SMOTE with the weighted random sampling technique, achieving better performance.

In the case of [36] we noticed the presence of infinity values in the dataset that the authors did not address. Additionally, handling of categorical variables was not specified. For testing their model, we replaced infinity and null values with the median. The study [36] is trained on 763,265 parameters, which requires more computational resources. In contrast, Our model is significantly more lightweight, with only 5052 trainable parameters, compared to 763,265 in [36], 13,792 in [11], and 308,435 in [35]. This reduction in parameter count theoretically reduces inference time. While we observed much faster inference empirically on our hardware setup, we acknowledge that these results are not derived under fully standardized benchmarking conditions. Therefore, instead of reporting specific multiples, we emphasize that our model is substantially more efficient in terms of computational cost and parameter footprint.

Finally, we implemented the model proposed by [11] training it on the full dataset. To handle infinity values, we replaced them with -1, and null records were deleted. The model was trained using the 69 suggested features, and the performance is reported in Table 10. Moreover, Our proposed model is computationally efficient, with fewer parameters than other models. The inference time and number of parameters are reported in Tables 10 and 12.

Overall, the experimental results highlight the effectiveness and efficiency of our proposed CNN-BiLSTM-Attention model. It achieves competitive or superior performance across multiple datasets while maintaining a low parameter count and demonstrating faster inference in our experiments. These experiments were conducted on a Windows 11 Pro workstation equipped with an Intel(R) Xeon(R) W-2155 CPU @ 3.30 GHz, 32 GB RAM, and an NVIDIA Quadro P4000 GPU (8 GB VRAM), using Python 3.9.13. While the observed computational speedups – especially against models with significantly higher parameter counts – are promising, we acknowledge that these were obtained within our controlled environment and may vary across different hardware and software configurations. Therefore, the inference time comparisons should be interpreted as indicative rather than absolute benchmarks. Future work will include standardized benchmarking protocols to provide more rigorous comparisons. Nonetheless, the model's compact size, strong accuracy, and robust generalization make it a practical choice for real-time deployment in DDoS detection systems, especially in resource-constrained environments such as edge devices or IoT gateways.

Table 10
Comparison of performance on CICDDoS2019 and Edge IIoT datasets.

Reference	Model	Dataset	Accuracy (%)	Time
[41]	CNN	CICDDoS2019	95.4 (B)	*
[11]	CNN	CICDDoS2019	99.97 (B), 94.57 (M)	27.01 μ s (Prediction time)
[42]	ML Models	CICDDoS2019	85 (M)	2.59 ms (Response time)
[20]	ID3-ML	CICDDoS2019	70 (M)	*
[43]	CNN	CICDDoS2019	95.12 (M)	*
[37]	CNN-AttBiLSTM	CICDDoS2019	95.00 (M)	*
[35]	DCNN-BiLSTM	Edge IIoT	99.64 (M)	219.29 s (Inference time)
Proposed Model	CNN-BiLSTM-AT	CICDDoS2019	100 (B), 99.78 (M)	441.61 s (Inference time)
		Edge IIoT	99.98 (B), 98.84 (M)	48.33 s (Inference time)

B: Binary Classification; M: Multiclassification.

Table 11
Performance of related models on different datasets (for reference only).

Reference	Model	Dataset	Accuracy (%)	Time
[44]	DBN	CICIDS-2017	99.37 (M)	*
[37]	CNN-AttBiLSTM	CICIDS-2017	95.67 (B)	*
[38]	CNN + BiLSTM	CICIDS-2017	99.83 (B), 99.81 (M)	*
[38]	CNN + BiLSTM	UNSW-NB15	94.22 (B), 82.91 (M)	*
[39]	Attention-based CNN-BiLSTM	InSDN	98.03 (M)	*

B: Binary Classification; M: Multiclassification. These models were tested on different datasets; comparisons are provided for contextual reference only.

Table 12
Performance Comparison of the Proposed Model with Existing Models.

Author	Model and Dataset	Parameters	Reported Metrics				Experimental Metrics			
			Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score
[40] (2021)	GB, CICDDoS2019	NA	99.97	98.9	99.99	99.44	99.97	98.9	99.99	99.44
[11](2021)	CNN, CICDDoS2019	13,792	99.97	99.99	99.98	99.98	99.91	99.97	99.93	99.95
[36] (2021)	CNN, CICDDoS2019	763,265	99.99	96.15	NA	97.3	99.89	99.84	99.95	99.89
[35] (2023)	DCNNBiLSTM, Edge IIoT	308,435	99.64	NA	NA	NA	94.78	94.78	94.77	99.78
Proposed Model	CNN-BiLSTM-AT,CICDDoS2019	5,052	99.78	99.77	99.77	99.70	99.78	99.77	99.77	99.70
	CNN-BiLSTM-AT,Edge IIoT	5,103	98.84	98.83	98.83	98.81	98.84	98.83	98.83	98.81

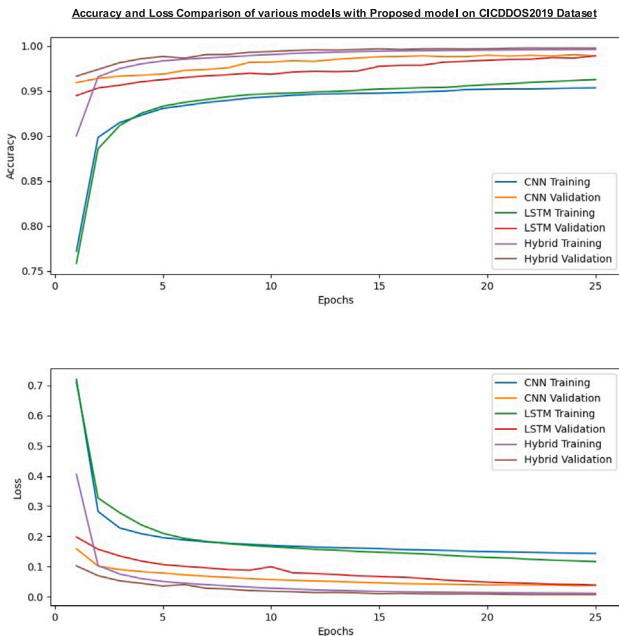


Fig. 16. Accuracy comparison of various models with proposed model on CICDDoS2019 dataset.

5.4.3 Real-world applicability

Our proposed CNN Bidirectional-LSTM model with an attention mechanism holds great promise for real-world DDoS intrusion detection applications. Key advantages include:

1. **High Accuracy and Robustness:** Achieving 99.78% accuracy, 99.77% precision, 99.77% F1 score, and 98.80% recall on CICDDoS2019, our model demonstrates reliable and confident intrusion detection.
2. **Low Computational Cost:** Despite outstanding performance, our model remains computationally efficient, enabling real-time intrusion detection.
3. **Handling Imbalanced Datasets:** Successfully managing highly imbalanced datasets like CICDDoS2019 and Edge IIoT showcases the model's adaptability and robustness.
4. **Industrial Deployment:** The efficiency and scalability of our model make it an ideal candidate for deployment in industrial environments, such as the Edge IIoT dataset. The timely detection of DDoS attacks in such settings is paramount to ensure operational integrity, data security, and optimal functioning of critical infrastructure.
5. **Network Security Applications:** Beyond industrial settings, our model holds immense potential for network security applications. It can be employed in diverse environments, including cloud-based services, online platforms, and critical infrastructures, safeguarding against DDoS threats effectively.

Therefore, our CNN Bidirectional-LSTM model with attention mechanism provides a practical and effective solution for bolstering network security and countering evolving cyber threats.

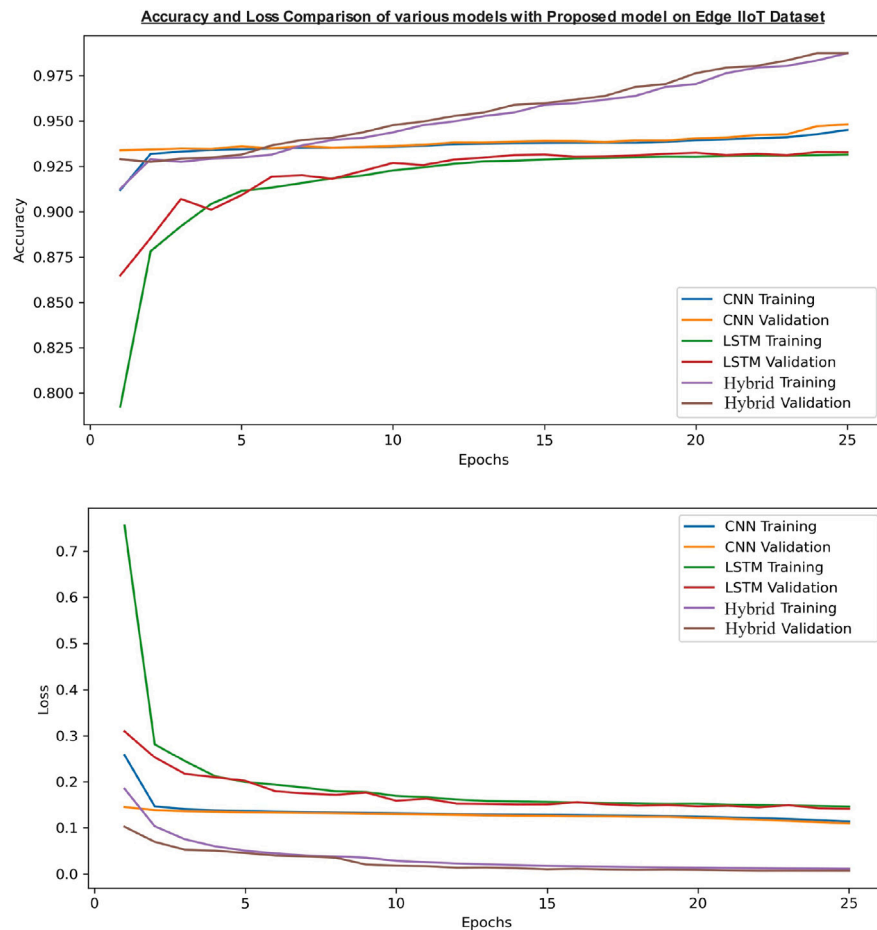


Fig. 17. Accuracy comparison of various models with proposed model on Edge IIoT dataset.

6 Conclusion

In this paper, we have proposed a novel hybrid deep learning model for intrusion detection in DDoS attacks. Our model integrates CNN, BiLSTM, and an attention mechanism to extract features from network traffic data and capture long and short-term dependencies. By leveraging deep learning techniques, our model effectively extracts higher-dimensional characteristics from network traffic data, enabling accurate multi-classification. Through rigorous evaluation on two benchmark datasets, CICDDoS2019 and Edge IIoT, we have demonstrated strong performance.

Our experimental results showcase the effectiveness of our proposed model, achieving a classification accuracy of 99.78% on the CICDDoS2019 dataset and 98.84% on the Edge IIoT dataset, with ROC AUC scores of 1.00 for CICDDoS2019 and 99.98% for Edge IIoT. Furthermore, comparisons with baseline models and existing studies underscore the potential of our CNN-BiLSTM model with an attention mechanism in accurately detecting and classifying network intrusions. In addition to its accuracy, our model also exhibits computational efficiency, making it a practical option for real-time DDoS detection, especially in resource-constrained environments. The broader implications of this work suggest potential applications in strengthening automated network defense systems.

However, several limitations warrant attention. First, some performance comparisons are made across different datasets and experimental setups, which may limit direct benchmarking accuracy. Second, all evaluations are based on single-run results; repeated trials and statistical reporting (e.g., variance or confidence intervals) would strengthen reproducibility. Additionally, while the model generalizes well across the tested datasets, its robustness against unseen or evolving attack

patterns remains to be validated. In future work, we plan to standardize evaluation pipelines, extend testing to include more recent or adversarial attack scenarios, and investigate online learning strategies to adapt to dynamic traffic patterns.

CRediT authorship contribution statement

Ashfaq Ahmad Najar: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S. Manohar Naik:** Supervision, Resources.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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